# ONR YIP technical proposal: panel review (v2)

Oct 29, 2026 · @Xinya Du

## Scope and conventions

This reviews printed pages 1–24 (Sections 1–6) of main-full.pdf at commit 4914b46 (Oct 29, 02:41 PT), whose main-full.tex is that of 6682ee9. My compile of the .tex matches the committed PDF; equations were read from the .tex.

- Page, figure and table numbers are the current ones; printed p. 1 is the third PDF page.
- Comments run C1–C61. C4–C13 are the ranked problems; C47–C61 are line edits. Sections follow your request, which has no item 5.
- Panel shorthand: M = the mathematician or statistician, A = the AI researcher, N = the Navy S&T reviewer.
- Every fix, cut and line edit marked "in the plan" is applied in one compiled file, main-full-fit20.tex (with its PDF and a diff that applies cleanly to 4914b46). Sections 1–6 then end on p. 20 with about 2 lines to spare, whether the three pilots keep their placeholders or take the expected-outcome sentences. See the Space plan at the end.
- I checked every claim about another paper against the paper itself and link it at first mention. Where I could not verify something, I say so.
- I read chatgpt_comment_v2.pdf only after my own pass, as a coverage check. Points that came from that check (C8′s numerical example, C22′s blame model, C34′s two example details, C58, C59) are marked "also in the ChatGPT review"; I verified each against the proposal.
- The file name claude_comment_v2.pdf is new. The similarly named claude_comments_v2.pdf is my corrected review of the earlier draft (029dff1).

## 1. Panel verdict

**Fund, once it fits 20 pages and the three placeholder results are resolved (C4, C5).** As submitted, Sections 1–6 end on p. 24, and three pilot results are orange placeholders; either one puts the proposal at risk before its science is read.

**Technical merit: Very Good** (up from Good for M in my v1 review). Panel summary: "A coherent, now properly credited framework. Each task controls one term of the error bound (1); the reliability model is Jaffe et al.'s dependent-classifier hierarchy with Kruskal-type identifiability; the choice of debaters and judge is a group partition mechanism with a clean one-half guarantee; and the training rewards rest on the classical noisy-label identity. The mathematics is correct. The propositions are design lemmas, and the hard results (how fast groups are detected, what coalitions gain, how much harm a colluder does before it is flagged) are honestly stated as the research. Direct evidence is good for RG1, a simulation and a real nine-agent pilot, and still missing for RG2 and RG3." A specialist will mark down awareness of the state of the art unless three novelty sentences are corrected (C7).

**Naval relevance: Very Good.** Panel summary: "The watch-officer example is concrete and runs through every task. Copied, stale and forged reports map credibly to maritime domain awareness, intelligence analysis, and AI test and evaluation, and the proposal answers both the 2026 ONR S&T Strategy's call for independent AI critique and the Mathematical Data Science call for principled methods. Testbed B reports the example's own measure, missed and false departure calls per 100 vessel questions. All data are public or synthetic and there is no Navy partner, which is normal for a YIP."

**Decision: fund**, on the two conditions above. If the page limit is missed, the NOFO does not say what happens to extra pages; any outcome (return, reading only to p. 20, a lower score) is bad, and stopping at p. 20 would drop Sections 3.6–6 (schedule, goals and benchmarks, Gold Standard Science, management, qualifications and responsibility), all of which the NOFO requires.

**The one change that most raises the odds**, beyond those two compliance fixes: run the Task 2.1 forged-tool pilot today and put its numbers where the orange sentence is on p. 15 (C30). It tests the proposal's most original mechanism, needs only a calculator, a small synthetic port table and one judge model, and gives RG2 its own evidence.

**Strengths not to edit away**

- **C1** (p. 7, eq. (1), rules R1–R5). The error bound with ε_cal: every task now owns a term that is measured separately, which is the structural answer to "ad hoc". Keep the bracket and the term-by-term ablations.
- **C2** (pp. 11–12, eq. (4), Proposition 3). The random halves: no group's ratings can change its own chance of a role, the chosen debater keeps half the best arguer's quality however many groups give the answer, and the uniform judge lottery is the best

possible against f colluders. This is now the proposal's cleanest mechanism-design result.

- **C3** (p. 19, "Checking the premise"; p. 21, the departure-call measure). Measuring how often the labels are wrong, stopping training on a topic when audits show the reference is wrong too often, and reporting departure calls per 100 vessel questions answer the colleague's (c) and (d) with procedures rather than promises.

## 2. Top problems

Four must-fix problems (C4–C7) decide whether the panel sees the proposal at its best. All fixes below are applied in the tested plan, which ends Section 6 on p. 20, except C5's three pilot sentences (left as placeholders; the expected-outcome versions also fit), C11 and C13.

| # | Problem | Priority | In the tested plan |
|---|---|---|---|
| C4 | Sections 1–6 run to p. 24, four pages over | Must-fix | Yes; the plan is the fix |
| C5 | Three pilot results are placeholders | Must-fix | Same length either way |
| C6 | Nine citations say more than their sources | Must-fix | Yes |
| C7 | Three novelty sentences a specialist can refute | Must-fix | Yes |
| C8 | RG3 summary sentences claim more than the math | Should-fix | Yes |
| C9 | H2 and G2 include a prediction that cannot fail | Should-fix | Yes |
| C10 | Recent work to credit where reviewers look | Should-fix | Yes |
| C11 | Open Philanthropy boundary still vague | Should-fix | No; same length, you supply the scope |
| C12 | A baseline a reviewer will name for H3 | Optional | Yes |
| C13 | Captions, tables and diagrams below 12 point | Optional | Kept small |

## C4. Sections 1–6 run to p. 24, four pages over the limit · must-fix

**Where:** the whole technical proposal; Section 6 ends about 171 lines past p. 20. The contents note says "occupies pages 1–24; the limit is 20 pages."

**Reviewer (program):** "The NOFO's limit is 20 pages for the technical proposal, and Management, PI Qualifications and Responsibility count inside it."

**Fix:** apply the tested plan (Space plan, end; main-full-fit20.tex, .pdf and .diff). It removes about 3.8 pages in four cuts, cheapest first (Layers 1, 2, 4 and 5 of the Space plan):

1. **Spacing only, −29 lines.** No extra space above run-in heads (3 pt now), 5-pt display skips, slightly tighter heading and float spacing, and no page-break penalty before section headings. No font, margin or wording changes.

2. **Text, −81 lines.** Duplicates and low-value sentences, section by section. No proposition, equation, hypothesis, metric or baseline is removed; the citations dropped are repeats, or PI papers that no task uses (they stay in the CV).
3. **Three exhibits, −29 lines.** Table 1 (its gaps are already the italic sentences of Section 2), Figure 4 (Figure 2's Task 1.1 box shows the same shared answers) and Figure 7 (its numbers are illustrative).
4. **Line pitch 13.8 pt instead of LaTeX's 14.5 pt, −43 lines.** One line, `\linespread{0.952}`. Word's "single" spacing for 12-pt Times New Roman is about 13.8 pt (measured), so the text stays single-spaced in the NOFO's sense. It is denser than the four funded examples, which I measured at 14.3–14.5 pt.

The fixes (C6–C10, C12) and the line edits add back about 9 lines. If you prefer a looser pitch: at 14.1 pt (`\linespread{0.97}`) dropping Figure 1 as well still leaves about 2 lines over; at 14.5 pt, dropping Figures 1 and 6 as well leaves about 4 lines over (both tested).

**Cut:** the plan is the cut. It also changes the contents note to "occupies pages 1–20, within the 20-page limit."

## C5. Three pilot results are still placeholders, and one measured result is true by construction · must-fix

**Where:** orange sentences on p. 15 (Task 2.1), p. 17 (Task 2.2) and p. 19 (Task 3); the Task 1.2 result on p. 13.

**Reviewer (all):** "61% to 1%, an AUC of 0.86 against 0.56, and 0.62 to 0.68 to 0.41 read as results. Unmeasured, they cannot be submitted; deleted with nothing in their place, they leave RG2 and RG3 without direct evidence."

**Fix:** run Task 2.1 today (C30). For Tasks 2.2 and 3, use the expected-outcome sentences already in the .tex comments under each pilot; they fit (tested). Compile the submission with `\finalbuildtrue`, which stops on any remaining `\pending`. On p. 13 (in the plan): "and not at all under our mechanism, whose chosen representatives reached 0.92 of the best arguer's quality" → "and, as Proposition 3(a) guarantees, not at all under our mechanism, whose representatives reached 0.92 of the best arguer's quality (the bound is 0.5)". The zero then reads as a check and the 0.92 as the finding. Also say in a few words how argument quality was measured; I could not tell from the text.

**Cut:** none needed; each replacement is about the same length (tested).

## C6. Nine citations say more than their sources · must-fix

**Where:** pp. 2, 5, 6, 13, 16, 19.

**Reviewer (A):** "A 78–90% figure that holds only for small open-weight video judges, an Ising model described as a shared latent factor, a sandbagging paper cited for debate, and peer prediction attributed to a paper that never mentions it. A reviewer who knows one of these discounts the rest."

**Fix** (all in the plan; the text gets shorter):

- **p. 2, Gap 3.** In Xu (2026), the 78–90% acceptance comes from three small open-weight judges of a video-generation agent; frontier judges were "essentially unmoved". Zhang et al. (2026) argue that agent harnesses should be disclosed and do not support the sentence. Replace the passage with: "Tools, retrieval, and memory let LLM agents check what their parameters cannot (Mialon et al., 2023), yet LLM judges still rule on the debaters' text alone. Such a judge may favor the more persuasive side even when one database query would settle the disputed step; small open-weight judges shown a log claiming that a tool call succeeded accepted 78–90% of a video agent's failed outputs, though frontier judges did not (Xu, 2026)." Those judges are a Chinese model family, so the sentence stays generic.
- **p. 5.** Balasubramanian et al. (2026) fit pairwise Ising couplings, not a shared latent factor: "with a shared latent factor" → "with pairwise (Ising) dependence". Their critique of Jaffe-style models is C7.
- **p. 6.** Wen et al. (2025) is disputed: an ICLR 2026 blog-track re-analysis found major bugs and, after correcting one experiment, failed "to find evidence that supports the original paper's claims" (Chandna, Fluri and Carroll). Delete "; and rewarding approval taught models to mislead evaluators (Wen et al., 2025)" and the approval clause in RG3's first challenge (p. 17). Shafayat, He and Kenton carry the point.
- **p. 6.** Yang et al. (2026) reward agreement with one designated peer's majority answer: "rewarding agreement with other models' majority relies on model diversity" → "rewarding agreement with a peer model's majority answer relies on cohort diversity".
- **pp. 5–6.** Hu et al. (2026b) count debate-helpful items (31 of 47 unanimous but wrong): "items that discussion helped" → "items that debate helped".
- **p. 13.** van der Weij et al. (2025) study underperformance on capability evaluations, with no debate: "by arguing badly on purpose (van der Weij et al., 2025)" → "by underperforming on purpose, as LLMs can on evaluations (van der Weij et al., 2025)".

- **p. 16.** Steinhardt et al. (2016) bound the number of trusted ratings, not a sequential test's audits. Drop that citation after "not on the number of decisions"; Ramdas et al. (2023) is cited in the preceding sentence.
- **p. 19.** Denisov-Blanch et al. (2026) test voting, confidence weighting and Surprisingly Popular, never peer prediction: "and peer prediction can then reward it" → "and agreement-based rewards then amplify it".

Everything else I checked is supported, including Kohli's "about 2 independent votes" from nine judges, Shi et al.'s 94%, AAAI-26's 22,977 papers, Kim et al.'s 60%, Khan et al.'s 76% and 88% against 48% and 60%, Shu's jump to near-unanimity, Fan et al.'s three to five times the tokens, the PRD numbers (Tables 3 and 8 of the TMLR paper) and HiPRAG's 27% to 2.3%.

**Cut:** none; the replacements are shorter.

### C7. Three novelty sentences a specialist can refute · must-fix

**Where:** p. 6 (end of "Discussion and debate among LLMs"); p. 12 ("Why this method"); Task 1.1, p. 11.

**Reviewer (A, M):**

1. "'We know of no debate protocol designed for many agents that share mistakes or collude.' AceMAD is a multi-agent debate protocol built for a majority that shares a misconception (Liu et al., 2026); credibility scores protect a multi-agent system from an adversarial majority (Ebrahimi et al., 2025); CP-WBFT weights agents against Byzantine faults (Zheng et al., 2025)."
2. "Proposition 3(a)'s group version exists: Divide-and-Rank keeps each connected component of the conflict graph on one side of its partition and is strategyproof (Xu et al., IJCAI 2019)."
3. "Balasubramanian et al., whom you cite, write that hierarchical dependence 'is poorly aligned with LLM-as-a-judge settings' because 'dependencies among LLM judges are rarely tree-structured or nested.' Task 1.1 builds on that hierarchy and never answers them."

Two more a reviewer may raise on Gap 3, which survive if you name them: a debate checker that verifies cited quotes mechanically (GAVEL, Findings of ACL 2026) and judges trained to run their own code (TIR-Judge, ICLR 2026). Neither re-runs the debaters' own calls.

**Fix** (in the plan):

- **p. 6.** Replace from "Others give agents tools" to the end of the paragraph with: "Others give agents tools (Jeong et al., 2026), call a tool-using verifier to break deadlocks (Li et al., 2026b), check cited quotes mechanically (Xu et al., 2026a), or train judges to run their own code (Xu et al., 2026b), but no judge re-runs the debaters' own calls. Recent protocols down-weight adversarial agents (Ebrahimi et al., 2025; Zheng et al., 2025) or a misled majority (Liu et al., 2026), *but none counts a shared mistake once or audits for collusion.*"
- **p. 12.** "extended from single agents to groups:" → "applied to groups as conflict-aware partitioning is in peer review (Xu et al., 2019):".
- **p. 11, Task 1.1 evaluation.** After "comparing with the classical reliability model, which is ours without shared factors" add ", and with a pairwise (Ising) dependence model that does not assume nested groups (Balasubramanian et al., 2026)". This turns their critique into a test you should win where copies of one source are nested by construction (testbed B's source trees), and lose gracefully elsewhere.
- The new bibliography entries are in main-full-fit20.tex; references do not count toward the limit.

**Cut:** the fixes C6–C10, C12 and the line edits add about 9 lines (Layer 3), paid in the Space plan.

## C8. RG3's summary sentences claim more than the math gives · should-fix

**Where:** p. 3 (central hypothesis; RG3 paragraph), p. 4 (1.3 (iv)), p. 17 (RG3 overview), pp. 18–19 ("Which debates count").

**Reviewer (M):** "The chance term of (5) is, as the text says, a baseline, so it changes no expected update. 'Rewards that agreement alone cannot raise' then rests only on leaving the agent's group out, and where the rest of the group shares a misconception, agreeing with it is rewarded, as p. 19 admits. And 'a message's expected credit is a positive multiple of how far it moved the judge toward the true answer' needs the decision's correctness not to depend on that message."

**Counterexample** (eq. (6); also in the ChatGPT review): yes-or-no debates, all settled by a re-run tool, 80% decided correctly. In the correct ones a message moves the judge 0.01 toward the decision; in the other 20% a misleading message moves it 0.40 toward the wrong decision. Mean credit before ω: $0.8 \times 0.01 + 0.2 \times 0.40 = +0.088$. Mean movement toward the truth: $0.8 \times 0.01 - 0.2 \times 0.40 = -0.072$.

**Fix** (in the plan):

- **p. 3, central hypothesis.** "through rewards that agreement alone cannot raise" → "through rewards that follow the true answer while the rest of the group is more often right than wrong".
- **p. 4, (iv).** "first-answer rewards that conformity cannot raise" → "first-answer rewards that echoing one's own group cannot raise".
- **p. 3, RG3 paragraph.** "occasional wrong decisions slow this training but do not reverse it," → "wrong decisions on yes-or-no questions slow this training but do not reverse it while they are a minority,". With more answers, one expected step can lower the right answer's probability (C20).
- **p. 17, RG3 overview.** "rewards for first answers that agreement alone cannot raise and that follow the true answer in expectation even though decisions are sometimes wrong, each weighted by how confident the rest of the group is" → "rewards for first answers that follow the true answer in expectation while the rest of the group is right more often than wrong, weighted by its confidence".
- **pp. 18–19.** Replace "But it rests on evidence rather than persuasion, so, while such decisions are right more often than wrong, a message's expected credit is a positive multiple of how far it moved the judge toward the true answer." with "But it rests on evidence rather than persuasion: for yes-or-no questions, if whether such a decision is right does not depend on the messages, then while such decisions are right more often than wrong, a message's expected credit is a positive multiple of how far it moved the judge toward the true answer; the testbeds' known answers test this condition."

**Cut:** paid in the Space plan.

## C9. H2 and goal G2 include a prediction that cannot fail · should-fix

**Where:** p. 13 (H2); p. 22 (Table 4, G2).

**Reviewer (M):** "'A single strategic rater gains nothing' is Proposition 3(a). A property of the mechanism is not a hypothesis, and listing it as one invites the question of what H2 risks."

**Fix** (in the plan): "H2 predicts that a single strategic rater gains nothing, coalitions gain less than our bounds allow, and impartiality costs less accuracy than debating adds." → "H2 predicts that coalitions across groups gain less than our bounds allow and that impartiality costs less accuracy than debating adds; a single group gains nothing by Proposition 3(a)." In Table 4, G2, delete "No gain for a single rater;".

**Cut:** none; shorter.

## C10. Recent work to credit where a reviewer will look · should-fix

**Where:** p. 10 (rule (3)); p. 18 (the identity and eq. (6)).

**Reviewer (A):** "Escalating from a judge to a stronger one below a calibrated confidence threshold is Trust or Escalate (Jung et al., ICLR 2025). Crediting each step by the change in the agent's own probability of the target is ΔBelief-RL (Auzina et al., ICML 2026). RL with noisy verifiers already corrects the update for the verifier's error rates (Cai et al., 2025)."

**Fix** (in the plan): p. 10, after "with a debate in place of rejection" add ", as LLM judges escalate to a stronger judge when unsure (Jung et al., 2025)". p. 18, "(Natarajan et al., 2013)" → "(Natarajan et al., 2013; Cai et al., 2025)", and after "(Wolpert and Tumer, 2001; Agogino and Tumer, 2008)" add ", as ΔBelief-RL does for one agent's own beliefs (Auzina et al., 2026)".

Optional, not in the plan: Motwani et al., Secret Collusion among AI Agents (arXiv:2402.07510), at red-team level 4; Dhull et al. on the expertise cost of partitioning in peer assessment (HCOMP 2022), beside Proposition 3(b); and Bara's concurrent argument that agents reading one source add evidence only up to a source-level ceiling (arXiv:2609.01873), the idea of Proposition 1.

**Cut:** paid in the Space plan.

## C11. The Open Philanthropy boundary is still one vague clause · should-fix

**Where:** p. 23, "Relation to other support"; p. 24, last bullet of Section 5.

**Reviewer (program):** "The PI holds an Open Philanthropy grant on control evaluations of AI debate, and Task 2.2 with its red team is a control evaluation of AI debate. 'This proposal funds aims those awards do not' does not say which aims."

**Fix** (same length; not in the plan): replace "the Open Philanthropy grant supports control evaluations of AI debate; the Amazon Research Award concerns process-guided fine-tuning. This proposal funds aims those awards do not" with "the Open Philanthropy grant supports [its scope in one clause] and does not fund [the aims it excludes: e.g., the reliability model, the impartial mechanism, the audit theory, RG3 training]; the Amazon Research Award concerns process-guided fine-tuning". I cannot verify the grant's scope, so the brackets are yours to fill.

## C12. A protocol a reviewer will name as H3's missing baseline · optional

**Where:** pp. 15–16, Task 2.1 evaluation.

**Reviewer (A):** "A collaborative protocol that locates and examines the point of disagreement reached 62.1% accuracy for non-expert judges against 49.2% for standard debate (Jiang et al., 2026). H3 compares debate only with open discussion."

**Fix:** "and replace debate with open discussion" → "and replace debate with open discussion or collaborative disagreement resolution (Jiang et al., 2026)". It is in the plan, which still fits.

### C13. Captions, tables and diagram text below 12 point · optional

**Where:** captions and Tables 1–4 at 10.95 pt; diagram text at 9.2 pt Helvetica (Figures 1–7).

**Reviewer (program):** "The NOFO says Times New Roman, 12-point." About 12% of the characters on pp. 1–24 are below 12 pt. Joshi's, Chen's and Yousefi's funded proposals have 4.5–9.3%, in figures, captions and tables, so small text there has passed before. Setting captions and tables at 12 pt costs about 10 lines (tested). Questions on technical requirements closed Oct 16; questions to the grants officer about submission close today, Oct 29, at 5:00 PM ET, and the NOFO warns that questions in the last two weeks may go unanswered. I would keep the small text and move on.

## 3. Math audit

Every proposition is correct under its stated assumptions, and rule (3) needs one it does not state (C17). A few sentences around them are wrong as written (C18, C20–C22; fixed by C8 and line edits C55–C57), and several known results are not credited (C7, C10). Proposition 2, Proposition 3(a)–(b), eq. (1) and the noisy-label identity carry real weight; the rest are honest design lemmas.

| Item | Correct? | New or known (credited?) | Design choice it justifies | Panel reading |
|---|---|---|---|---|
| Eq. (1), p. 7 | Yes | Standard selective-prediction split (n/a) | Task-to-term map, ablations, τ | Load-bearing |
| Prop. 1, p. 9 | Yes | Immediate from Jaffe et al. (yes) | Groups enter through one shared answer | Decorative, honest |

| Item | Correct? | New or known (credited?) | Design choice it justifies | Panel reading |
|---|---|---|---|---|
| Prop. 2, pp. 9–10 | Yes | Corollary of Allman et al. 2009 (yes) | At least three independent groups; H1 | Load-bearing |
| Rule (3), p. 10 | Yes, if ε_D does not vary with π_q | Chow 1970 (yes); Jung et al. 2025 (no) | τ from cost and stakes | Load-bearing |
| Prop. 3(a), p. 12 | Yes | Xu et al. 2019 for groups (no) | Ratings only from the other half | Load-bearing |
| Prop. 3(b) | Yes, if raters share one ranking | Alon et al.'s partition (yes); Dhull et al. on its price (no) | Random halves | Load-bearing |
| Prop. 3(c) | Yes, trivially | Standard minimax | Uniform judge lottery | Decorative |
| Prop. 4 and the ε_step bound, pp. 13–14 | Yes | Irving et al. (yes) | Step disputes, re-running, judge training | Load-bearing with the bound |
| Identity for (5), p. 18 | Yes, for yes-or-no | Natarajan et al. (yes); Cai et al. (no) | Leave-group-out reference; ω after GRPO | Load-bearing; chance term decorative |
| Eq. (6), p. 18 | Accounting yes; truth claim needs a condition | ΔBelief-RL (no) | Credit only in tool-settled debates | Load-bearing once conditioned |
| Harm rate, p. 16 | Plausible target, not shown | Howard et al. for the slack (yes) | G4 | The research |

## C14. Eq. (1), p. 7

- **Correct?** Yes. On $\{\pi \geqslant \tau\}$: $E[P(\hat{y} \neq y \mid \pi)\, 1\{\pi \geqslant \tau\}] \leqslant E[(1-\pi)\, 1\{\pi \geqslant \tau\}] + \varepsilon_{cal} \leqslant (1-\tau)\, P(\pi \geqslant \tau) + \varepsilon_{cal}$. On $\{\pi < \tau\}$: a union bound over "neither debated answer is right", "a dishonest agent steers" and "an unsteered debate ends wrong", with the bracketed terms read as probabilities given a debate. $\varepsilon_{cal}$ removes the calibration assumption my v1 review flagged.
- **Counterexample:** none to the bound. The terms are not independent levers (Task 3 moves all of them, and colluders feed both $\varepsilon_{cal}$ and $\varepsilon_{role}$); the text says both.
- **Realistic for LLM agents?** Yes, because $\varepsilon_{cal}$ is estimable without bias from the random human audits at rate $\rho$.
- **Justifies** the task-to-term map and the per-term ablations of Section 3.5. Load-bearing.

## C15. Proposition 1, p. 9

- **Correct?** Yes. The likelihood ratio $[(1-\lambda)\Lambda + \lambda] / [(1-\lambda) + \lambda\Lambda]$ has derivative of sign $(1-\lambda)^2 - \lambda^2$, positive for $\lambda < 1/2$, and runs from $\lambda/(1-\lambda)$ to $(1-\lambda)/\lambda$. The $|Y|$-answer bound also holds, for $\lambda \leqslant (|Y|-1)/|Y|$: a ratio of two linear functions of the members' answer probabilities is extreme at a vertex.
- **Realistic?** One hidden shared answer per question, members independent given it. Right for copies of one stale source, which testbed B builds; doubtful for agents sharing an underlying model, whose errors are correlated but differ question by question. That is Balasubramanian et al.'s critique (C7); the Ising baseline in the plan tests it.
- **New?** Immediate from Jaffe et al.'s model, which is credited. Bara (2026) makes the same evidence-ceiling argument informally.
- **Panel reading:** decorative but honest. The content is in Proposition 2 and the partition search.

## C16. Proposition 2, pp. 9–10

- **Correct?** Yes. Three conditionally independent views of full column rank give Kruskal ranks summing to $3|Y| \geqslant 2|Y| + 2$, so the decomposition is unique up to a common relabeling (Allman et al. 2009). The parenthetical sufficient condition checks: $\kappa_g = (1-\lambda_g)I + \lambda_g N_g$ is column-diagonally dominant when $\lambda_g < 1/2$ (each column of $\lambda_g N_g$ sums to $\lambda_g < 1 - \lambda_g$), hence invertible, and a column-wise Kronecker product of the members' confusion matrices has full column rank if one factor does. The two-agent

example is right: only $(2p_a - 1)(2p_b - 1)$ is identified. With three agents, $(2p_a - 1)^2 = c_{ab} c_{ac} / c_{bc}$, where $c_{ab} = (2p_a - 1)(2p_b - 1)$, up to one global sign, provided no agent sits at exactly 50%.

- **v1 gaps closed:** "jointly with two other groups" fixes the co-answering condition, and development questions fix the labels.
- **Realistic?** Independence across groups given the truth is the weak point; Kim et al. (2025) find errors correlated even across providers, so the partition must absorb cross-family correlation. Colluders who coordinate across groups break it too, and flags merge them.
- **Load-bearing:** it is the only proposition that predicts a failure (fewer than three independent groups), and H1 tests it.

## C17. Rule (3), p. 10

- **Correct?** Yes, given a calibrated $\pi_q$ and an $\varepsilon_D$ that does not depend on $\pi_q$.
- **Counterexample to "exactly when":** hard questions likely have both low $\pi_q$ and high $\varepsilon_D$. Let $\varepsilon_D = 0.5$ when $\pi < 0.6$, $\varepsilon_D = 0.05$ otherwise, and $c/L = 0.05$. Debating is worth it for $\pi < 0.45$ and for $0.6 \leqslant \pi < 0.9$, which is not a threshold. A pooled $\varepsilon_D$ of 0.2 gives $\tau = 0.75$: it debates $\pi$ in $[0.45, 0.6)$, where debate is worse than standing, and skips $[0.75, 0.9)$.
- **Fix** (in the plan): "separately for disputes a tool can settle" → "separately for disputes a tool can settle and by bins of $\pi_q$".
- **New?** Chow (1970), credited; cite Jung et al. (2025) (C10). Load-bearing: it sets $\tau$.

## C18. Proposition 3, p. 12

- **(a) Correct.** The judge, the halves, the fallback and the weights ignore current ratings, and (4) can choose a member of $g$ only for the answer whose half holds $g$, from the other half's ratings. One nearby sentence is wrong: "An agent either rates or may be chosen, never both." When $C_y$ is empty, the fallback can draw an agent of $y$'s half that rates for the other answer. Impartiality still holds because the fallback ignores ratings; the plan deletes the sentence.
- **(a) New?** The group version is Xu et al. (2019) (C7).
- **(b) Correct.** Swapping the halves preserves the coins' law and the redraw condition, so $g(a^*_y)$ lands in $y$'s half with probability exactly 1/2; then $a^*_y \in C_y$, $V_y$ is nonempty, and (4) picks an agent at least as good. The redraw always succeeds once two groups

besides the judge′s remain: two debated answers, each given by two or more groups, can always be split across the halves.

- **(b) Realistic?** "Each rater′s ratings rank agents by argument quality Q" means all raters share one ranking. LLM raters disagree and favor similar models (Panickssery et al. 2024). With noisy raters (b) no longer holds as stated and the price must be measured, which you do; report the 0.92 against both 1/2 and plain weighted ranking.
- **(c) Correct and immediate** (the top f agents of any lottery carry at least f/n_q). Decorative; it can stay as one clause.

## C19. Proposition 4 and the ε_step bound, pp. 13–14

- **Correct.** A union bound over (i), (ii) and k rulings gives ε_step ⩽ P(not (i)) + P(not (ii)) + kε_J.
- **Realistic?** (ii) is the obfuscated-arguments problem: the honest side must find a false step in an argument the other side can make long, which prover–estimator debate (Brown-Cohen et al. 2025, cited for recursion) addresses. (iii) holds roughly for tool-checkable steps and is what judge training targets. (i) fails for true but selective facts, which the right of reply addresses.
- **Evidence quoted:** Ling et al.′s 69% against 52% compares one-shot step-by-step prompts with two-shot whole-chain prompts; their zero-shot step-by-step accuracy is 53%. Part of the gain is the worked example. Line edit C54 says so in six words.
- **Load-bearing** with the bound: each harness part lowers one measured term.

## C20. The identity behind eq. (5), pp. 17–18

- **Correct** for yes-or-no questions with a reference independent of the agent′s answer: E 1{x = D} = η + (1 − 2η) 1{x = y}, so the expected policy gradient is (1 − 2η) times the true-label one. ω_q applied after GRPO′s per-question normalization keeps the weighting (fixed since v1).
- **Counterexample beyond yes-or-no:** three answers; the reference gives (y, $w_1$, $w_2$) with probabilities (0.55, 0.40, 0.05); a softmax policy puts (0.01, 0.98, 0.01) on them. The expected REINFORCE step raises $w_1$′s logit by 0.00196 and y′s by 0.00152 per unit step, so π(y) falls (by $3.8 \times 10^{-6}$). Section 3.4 restricts the identity to yes-or-no; the p. 3 summary did not (C8).
- **The chance term** is a baseline, as the text says: it zero-centers the score but changes no expected update, so it does not by itself stop conformity. Keep it for scoring; claim nothing more for it (C8).

- **Weighting:** multiplying by 1 − 2η̂ down-weights noisy questions, while the unbiased correction divides by it (Natarajan's backward correction). Multiplying is a sound variance choice; one clause would say so to a statistician.
- **Realistic?** The reference stays independent of the agent while the partition is right, and erodes as training makes agents alike, which you measure. η_q < 1/2 fails on shared misconceptions; the topic stop rule and held-out checks are the guard.
- **Load-bearing:** this is the answer to the colleague's (c) for first answers.

## C21. Eq. (6), p. 18

- **Correct as accounting:** the credits sum to ω_q[P_J(D | m_1:T) − P_J(D | no messages)], and the text rightly says policy invariance does not carry over.
- **The truth claim needs a condition** (C8's counterexample). Inside tool-settled debates the condition is plausible for messages other than the tool evidence. The tool evidence earns positive credit whether the tool is right or wrong, which is fine in expectation if the tool's error does not depend on the message.
- **New?** ΔBelief-RL credits steps by the agent's own belief change (C10); using an impartially drawn judge's belief and crediting only tool-settled debates is the new part.

## C22. Task 2.2's rates, p. 16

- **"Cannot be dodged":** correct. Audits drawn after the decision are independent of what the colluder infers, so its first n steered decisions all escape with probability (1 − ρ)^n.
- **√(n log log n / ρ):** plausible, and framed as a target. The heuristic: about ρn audits; an anytime-valid test must allow an excess of order √(ρn log log(ρn)) wrong audited decisions; scaled by 1/ρ to all n decisions, that is √(n log log n / ρ). "When they rarely do, it is a bounded number" is loose: with no honest errors and correct audits, a colluder steers about 1/ρ decisions before the first audited one (line edit C56).
- **"Which random pairings make identifiable"** holds for an additive model of agent effects, not for pair effects such as two colluders who fail only together (also in the ChatGPT review). The plan says "an additive model"; pair effects belong in the coalition analysis.
- **α control** holds against the predicted honest error rate. If the prediction is biased, say for an honest agent often paired with colluders, false flags can exceed α; the partner effects the text mentions are the guard.

# 4. The "ad hoc" test

The program officer's "ad hoc" comment is now answered for every task: each method follows from a stated model or mechanism, and what remains heuristic is named as research. The largest change since my v1 review is Task 1.2.

## C23. Task by task, against the white paper (v4)

| Task | White paper | Proposal now | Verdict | Still heuristic |
|---|---|---|---|---|
| 1.1 | Dawid–Skene "with shared error factors"; agents giving the same unusual answers share a factor | Model (2), Propositions 1–2, rule (3), EM with a partition search | Principled | Partition search, and audit flags that "enter as a prior"; the partition question is named under "What remains open" |
| 1.2 | "Partition idea"; top-rated member of a randomly drawn group | Random halves, eq. (4), Proposition 3(a)–(c) | Principled (biggest gain) | Coalitions across groups; named as research |
| 2.1 | Harness; walk to the first step where the sides differ; judge re-runs tool evidence | Proposition 4 with the ε_step bound; each harness part tied to one term; trained judges | Principled in structure | False-step search and the right of reply, each assigned to a term |
| 2.2 | Planted debates; random human audits; flag agents worse on audited than planted cases | Random inspection, anytime-valid test, harm target | Principled design | Blame model and α under partner effects (C22) |

| Task | White paper | Proposal now | Verdict | Still heuristic |
|---|---|---|---|---|
| 3 | Each message rewarded for moving the group toward the decision; tool-backed decisions weigh more | Leave-group-out reference with the noisy-label identity; credit only in tool-settled debates; topic stop rule | First answers principled; messages once C8′s condition is stated | The condition behind eq. (6) |

## C24. The colleague's questions

| Question | Status | Where |
|---|---|---|
| (a) Reliability without ground truth, with correlated errors and dishonest agents | Answered | Propositions 1–2, pp. 9–10; ε_cal, p. 7; colluders across groups and flags, p. 10; harm target, p. 16 |
| (b) Why trust the judge; independent is not reliable | Answered | "The judge need not be wiser than the debaters", p. 13; Proposition 4 and kε_J, pp. 13–14; judge training and pool, p. 15; audits, p. 16 |
| (c) Wrong group decisions reinforcing bad behavior | Answered for first answers; for messages once C8 states its condition | Identity and leave-group-out, p. 18; credit only in tool-settled debates, p. 18; topic stop rule, p. 19 |
| (d) Concrete metrics | Answered | Per-task metrics and H1–H5; Table 4 with numbers (G1: 5 points; G5: half the RLVR gain); departure calls per 100 vessel questions, p. 21 |

Question (e), the Navy connection, is Section 7 below.

### C25. The program officer's six questions

Each has a home: the problem (p. 1), why it matters (pp. 1–3), today's methods and limits (Section 2), the approach and why it should work (Sections 1.2–1.3 and 3), limitations (p. 21), and impact (pp. 3–5, 21). The weakest is "what is new" until C7's three sentences are fixed.

### C26. Changed or dropped since the white paper

- **Changed for the better:** Task 1.2 went from "the top-rated member of a randomly drawn group" to random halves with a one-half guarantee; "no clear leader" became an expected-loss threshold; "tool-backed decisions weigh more" became crediting messages "only in debates whose disputed step the judge settled with reproduced tool evidence", which Figure 2 now matches.
- **Changed:** "walk through their arguments to the first step on which they differ" became "each names the first step of the other's argument that it disputes", so the judge rules twice, hence k = 2 in the bound.
- **Dropped, and fine:** "keeps working after the agents outgrow human judges" left p. 3 in the latest commits; it survives as "can scale to AI whose answers people cannot easily evaluate" (p. 21).
- **Kept:** one question with separate tasks, the vessel example, three testbeds, the red team, and the baselines, now with a tool-using single judge.

### C27. My v1 points, now

Adopted: credit to Jaffe et al. and Chow; $\varepsilon_{cal}$ in (1); random halves with all three conditions (raters only from the other half, both halves nonempty, debaters from different groups); $k\varepsilon_J$; $\omega$ after GRPO's normalization; yes-or-no scope for the identity; message credit only in tool-settled debates; numeric G1 and G5; the testbed B measure; PRD's uptake in Section 5. Still open: the Xu (2026) overstatement (C6), the Open Philanthropy boundary (C11), Jiang et al. as a baseline (C12), and Cai et al. (C10).

## 6. Pilots

Run the Task 2.1 pilot today; run Task 2.2's discriminator test only if time remains; restate Task 3 as an expected outcome. The two measured pilots stand.

| # | Pilot (page) | Status | Would the result convince? | Before Oct 30? |
|---|---|---|---|---|
| C28 | 1.1: shared hint, nine agents, 300 GSM8K problems (pp. 10–11) | Measured | Yes, as a controlled demonstration | Done |
| C29 | 1.2: strategic rater on PeerEval (p. 13) | Measured | The 0.92 yes; "not at all" is Proposition 3(a) | Done; reword (C5) |
| C30 | 2.1: forged tool outputs, 200 disputes (p. 15) | Placeholder | Yes; tests the central novelty | Yes, first |
| C31 | 2.2: discriminator AUC; flagging simulation (p. 17) | Placeholder | The AUC yes; the simulation is a design calculation | AUC only, if time |
| C32 | 3: GRPO on GSM8K, 1.5B model (p. 19) | Placeholder | Yes, if the majority reward is self-referential | No |

## C28. Task 1.1, measured

Majority vote falling from 89% to 70% while the shared-factor model holds 87% and finds the four hinted agents as one group in 9 of 10 runs is direct evidence for Propositions 1–2. A skeptic will ask about shared errors with no injected hint, and about the two-group case where Proposition 2 predicts failure. H1 covers both; leave it.

## C29. Task 1.2, measured

The informative number is 0.92 of the best arguer's quality against a guaranteed 0.5. "Not at all under our mechanism" is true by construction, so present it as a check (C5), and say how quality was measured.

## C30. Task 2.1, placeholder: run it first

This pilot tests the most original mechanism and gives Gap 3 its own evidence. A half-day setup:

1. 200 disputes: 100 arithmetic steps (calculator) and 100 port-status steps (a synthetic 50-row port table with timestamps).
2. The wrong side attaches a forged tool output; the right side attaches the genuine one.

3. One judge model rules twice: from the transcript alone, and after re-running both calls.
4. Report forged-output acceptance and accuracy in each condition, with confidence intervals.

Re-running defeats a forgery by construction, so the informative numbers are the transcript-only acceptance rate and the residual error after re-running (a wrong query, a misread output). If time allows, add the two controls in the commented sentence, a stale record and a true but selective fact, where re-running should not help. Use a non-Chinese judge model, or describe the model generically. If it is not run, use the expected-outcome sentence already in the .tex comments.

### C31. Task 2.2, placeholder

The AUC contrast between templated and edited planted cases is a real, cheap experiment: a few hundred real questions, their templated and edited planted versions, one strong model as the discriminator. The flagging "simulation" follows from the test's own definition; if you give a number, call it a design calculation, not a pilot result. If neither is run, use the commented sentence (it fits).

### C32. Task 3, placeholder

Three GRPO runs plus reference answers are too much for tomorrow; use the commented sentence. If runs already exist, say whether the majority reward includes the trained model's own samples: the collapse in Shafayat et al. (2025) needs that self-reference.

### C33. Order of work

Task 2.1 first, then Task 2.2's AUC if time remains, then compile with `\finalbuildtrue`. Each measured sentence replaces its orange one in about the same space.

## 7. Naval relevance and skim test

A Navy reader will find the example credible and the work clearly basic research with broad use. Two wording points in the example are worth a minute; nothing here needs new content.

## C34. Is the watch-officer example credible? (p. 1; Figure 2, p. 2)

Yes. Maritime operations centers do fuse schedules, port reports, position tracks and imagery, and the proposal cites spoofed position broadcasts and untracked vessels. A Navy reader may push on two details (both also in the ChatGPT review):

- **Time.** The 09:10 port report shows V at berth at 09:10, not at whatever later time the officer asks. Fix the question's time in Figure 2 (C61).
- **Who acts.** A vote does not send a patrol; the officer does (C48).

The answer the officer needs in minutes is covered in principle by the debate cost c in rule (3); one clause saying c includes delay would close it, if you find the space.

## C35. Basic research with broad use, or a narrow demo?

Basic research with broad use. The methods are general, the testbeds span math and code, multi-document questions and peer review, and Section 1.4 names five Navy needs. The risk runs the other way: the Navy appears only in synthetic scenarios, which the departure-call measure partly offsets.

## C36. Does the title match?

Broadly. "Trustworthy Collective Reasoning and Evaluation for AI Agents" is generic and omits what is distinctive (shared mistakes, collusion, evidence the judge re-runs), but the program officer saw it on the white paper. Keep it.

## C37. Skim test: title, page 1 and Figure 1

The three sentences a busy panelist carries into the discussion:

1. "She wants a group of AI agents to give one trustworthy answer when some copy the same stale source and one is compromised, like five tools disagreeing on whether a ship has left port."
2. "Her pipeline counts copies once, debates only the unclear questions before an impartially drawn judge who re-runs the evidence, audits a few random decisions for collusion, and retrains the agents on the outcomes."
3. "Principled and Navy-relevant; the mathematics assembles known tools carefully, and the open bounds are the research."

Where a non-specialist first gets lost: Figure 2's Task 1.2 box, "Groups split at random: {A3} and {A4} may debate; {A1, A2} and {A5} rate them." The halves never appear, so the reader cannot see why A1 rates A3. C50 names the halves. The Task 1.1 box's "share an error

factor″ is jargon at first sight (C49).

## 8. Against the funded examples

Your framing, open problems and naval relevance now match or beat Joshi′s and Chen′s. The gaps a panel would notice are evidence for RG2 and RG3, length discipline, and density.

| # | What the funded proposals do | Where yours falls short | Worth it within 20 pages? How |
|---|---|---|---|
| C38 | Joshi builds each task on her own published estimator or bound and shows it | RG1 has a simulation and a real pilot; RG2 and RG3 rest on placeholders | Yes: run Task 2.1 (C30) |
| C39 | Joshi′s PI section is about half a page | Yours is 38 lines, close to a page, though the CV is scored separately (NOFO criterion 3) | Yes: the plan trims Section 5 to 30 lines |
| C40 | All four funded examples use LaTeX-default line pitch, which I measured at 14.3–14.5 pt | The plan′s 13.8 pt is denser | Partly: offset by dropping three exhibits; a 14.1-pt pitch needs Figure 1 dropped and 2 more lines cut (C4) |
| C41 | Joshi states the rates she will prove | Your one rate, $\sqrt{(n \log \log n / \rho)}$, is a target, and Proposition 2 gives no sample size | No new theorem before Friday; G4 names the target |
| C42 | Yousefi ties the work to named ONR program emphases | No gap: you quote the 2026 ONR S&T Strategy and the Mathematical Data Science call, more specific than Joshi or Chen | Keep as is |

# 9. Overlap and NOFO compliance

No task duplicates the CAREER award or the two pending NSF proposals; the overlap a reviewer will probe is the Open Philanthropy grant (C11). The compliance failures are the page count and the placeholders; everything else passes or is a judgment call.

## Overlap

| # | Proposal | What it covers | Overlap with this proposal | Boundary drawn? |
|---|---|---|---|---|
| C43 | NSF CAREER (funded): consistent event graphs from long documents | Event extraction and event knowledge graphs | Testbed B reuses MEQA-style event structures | Yes, p. 23 |
| C44 | NSF "Explainable Methods for Natural Language Reasoning" (pending) | Deductive, abductive and inductive reasoning; the MEQA dataset; explanation evaluation | MEQA only; no debate, judges, collusion or multi-agent RL | Not needed |
| C45 | NSF "Automating Scientific Knowledge Discovery through Retrieval, Generation, and Validation" (pending) | Hypothesis generation; reward models trained on OpenReview ratings; LLM evaluation | Testbed C also uses OpenReview papers and re-runs released code | Only by the generic sentence on p. 23; one clause would do: "testbed C uses OpenReview papers only as questions for collective evaluation" |

## C46. Compliance checklist (NOFO N0001426SF004)

| Requirement | Status |
|---|---|
| Technical proposal at most 20 pages | Fails: 24 pages (C4). Passes in the plan |
| No placeholder results | Fails: three (C5). `\finalbuildtrue` guards the final build |
| Contents note | Says "pages 1–24"; the plan makes it "pages 1–20, within the 20-page limit" (C47) |
| Times New Roman, 12 point | Body in Times (Nimbus Roman, the standard LaTeX clone, as in all four funded examples); captions and tables 10.95 pt, diagrams 9.2 pt Helvetica (C13) |
| Single-spaced, 1-inch margins | Passes; the plan's 13.8-pt pitch equals Word's single spacing for 12-pt Times New Roman (C4) |
| ONR S&T Focus Area | Passes: Artificial Intelligence and Autonomy (Intelligent and Collaborative Agents; Adversarial AI & Autonomy), on the cover and p. 4 |
| Gold Standard Science (EO 14303), with past success | Passes, p. 22: MEQA, DSBench and LMR-Bench released with data and code |
| Management, PI Qualifications, Responsibility inside the limit | Passes once the limit is met; all six Responsibility items are on p. 24 |
| Cover page | Passes: "Technical Proposal", N0001426SF004 (the announcement's number; the NOFO's cover instructions misprint it as N0001426SF0004), contacts, period |
| No Chinese model named | Passes on pp. 1–24; DeepSeekMath appears only in the reference list for GRPO, which you allow. The C6 fix keeps Xu (2026)'s judges unnamed |
| Eligibility statement | Passes, p. 24 |

| Requirement | Status |
|---|---|
| CV cross-check | The appended CV lists the same area-chair roles as p. 23 (ICML 2027, NeurIPS 2026, ICLR 2025 and 2027) |

## 10. Line edits

Fifteen edits: seven on pp. 1–5 and one on the contents page. All fifteen are in the tested plan.

- **C47** (contents note, p. ii). "occupies pages 1–24; the limit is 20 pages." → "occupies pages 1–20, within the 20-page limit." (after the cuts)
- **C48** (p. 1). "A majority vote picks the wrong answer and sends the patrol to sea after a ship still at berth;" → "A majority vote picks the wrong answer, and an officer who trusts it sends the patrol to sea after a ship still at berth;"
- **C49** (Figure 2, Task 1.1 box). "so they share an error factor and count about once." → "so the model treats them as one source and counts them about once."
- **C50** (Figure 2, Task 1.2 box). "Groups split at random: {A3} and {A4} may debate; {A1, A2} and {A5} rate them. A3 argues yes, A4 no." → "Random halves: {A3, A5}, {A1, A2, A4}. A3 argues yes, rated by A1, A2; A4 argues no, rated by A5." (This matches Figure 5's halves.)
- **C51** (p. 3). Merge the two paragraphs "These gaps matter more as AI evaluates AI at scale …" and "Navy staff who must act …" into: "These gaps matter more as AI evaluates AI at scale: AI judges supply the rewards that train models (Bai et al., 2022; Lee et al., 2024), and AAAI-26 gave an AI-generated review to each of the 22,977 main-track papers that entered full review (Biswas et al., 2026). As AI surpasses people at more tasks, oversight will rely on AI evaluating AI, with people auditing a small part of the work (Bowman et al., 2022; Burns et al., 2024). Navy staff who must act on one answer from several AI tools already face all four gaps (Section 1.4)." Section 1.4 already lists the Navy stressors.
- **C52** (p. 4, item (i)). "keeps at least half the best debater's quality" → "keeps, in expectation, at least half the best debater's quality" (Proposition 3(b) is a statement about E[Q]).
- **C53** (p. 10). "separately for disputes a tool can settle." → "separately for disputes a tool can settle and by bins of π_q." (C17)

- **C54** (p. 14). "raised ChatGPT's verification accuracy to 69%, from 52% (about chance) when it checked whole chains at once" → "raised ChatGPT's verification accuracy to 69%, from 52% when it checked whole chains at once, with one or two worked examples" (C19)
- **C55** (p. 16). "Blame is assigned by a model of each agent's effect" → "Blame is assigned by an additive model of each agent's effect" (C22)
- **C56** (pp. 16–17). "when they rarely do, it is a bounded number." → "when they rarely do, it is of order 1/ρ, however long the stream." (C22)
- **C57** (p. 12). Delete "An agent either rates or may be chosen, never both." The fallback draw can make it false (C18).
- **C58** (p. 19). "(standard RLVR) as an upper bound" → "(standard RLVR) as a reference". Training on known answers is a benchmark, not a bound on what label-free training can reach (also in the ChatGPT review).
- **C59** (p. 19). "Every question has a known answer for scoring, kept apart from the planted cases;" → "Every question has a known answer for scoring (in testbed C, only claims with a reproducibility report or a planted error are scored), kept apart from the planted cases;". Without it, "every question" is not true of testbed C (also in the ChatGPT review).
- **C60** (p. 5, if Table 1 goes). "We build on six lines of work and note the gap each leaves (Table 1)." → "We build on six lines of work; the gap each leaves is in italics."
- **C61** (Figure 2, question box). "Has cargo vessel V left port? (In fact, it has not.)" → "Has cargo vessel V left port by 09:10? (It has not.)" The question then matches the 09:10 port report (C34).

## Space plan (compile-tested)

Applied together to 4914b46, the edits below end Section 6 on p. 20 with about 2 lines to spare. Files: main-full-fit20.tex, main-full-fit20.pdf and main-full-fit20.diff; `git apply --check` passes on 4914b46. Other sessions are still committing, so if a hunk fails later, re-apply it by hand from the exact text.

Lines are 12-pt lines at today's 14.5-pt pitch, measured by compiling after each layer. Savings depend on where floats land, so read single numbers as approximate.

| Layer | What changes | Lines | Past p. 20 after it |
|---|---|---|---|
| Start | Commit 4914b46 | | +171.3 |
| 1. Spacing | No extra space above run-in heads (`\hd`, 3 pt now); 5-pt display skips; section spacing 6/2 pt, subsection 5/1 pt, subsubsection 3/0 pt; float separation 6 pt, in-text floats 5 pt; caption skip 5 pt; no page-break penalty before sections | −29.2 | +142.1 |
| 2. Text | The trims listed below | −81.3 | +60.8 |
| 3. Fixes and credits | C6–C10, C12, the final-decision baseline kept in Task 3′s evaluation, and the line edits (C51, C57 and C8′s p. 17 bullet are counted in Layer 2, C60 in Layer 4); six references no longer cited removed, and DynaDebate relabeled Li et al. (2026) | +8.6 | +69.4 |
| 4. Three exhibits | Table 1, Figure 4, Figure 7; Figure 5 moved to the Proposition 3 paragraph so its wrap does not run onto p. 11 | −28.6 | +40.8 |
| 5. Line pitch | `\linespread{0.952}`: 13.8 pt, Word′s single spacing for 12-pt Times New Roman | −42.8 | 2.0 lines to spare |

**Layer 2, by section.** Each item is an exact replacement in the diff.

- **Section 1 (about −2):** C51′s merged paragraph; RG3 paragraph shortened; the last two Navy bullets merged into "Test and evaluation, and other uses"; "Path to transition" in one sentence.
- **Section 2 (about −4):** the Nitzan–Paroush sentence (kept in Task 1.1); the Flanigan clause (kept in Task 1.2); "LLM judges are standard for evaluating models"; the Choi sentence tightened.
- **RG1 and Tasks 1.1–1.2 (about −17):** the repeated "LLM errors are correlated" citations; overview bullets to one line each; the Task 1.1 opening tightened; the vessel sentence after rule (3) (Figure 2 shows it); "shared sources serve only as clues"; "the halves are random sets of groups"; the Figure 5 walk-through (its caption and Figure 2 give it); the few-groups sentence shortened; C57′s sentence.

- **RG2 and Tasks 2.1–2.2 (about −13):** collusion citations repeated from Section 2; the five harness bullets folded into one paragraph (Figure 6 shows the parts); the vessel sentence; "A pool of judges" in one sentence; the PRD discussion result in Task 2.1's preliminary results (it supports Gap 2, not the harness); the Khan clause in judge training; the COMET paragraph tightened; "Proposition 4 is the classical argument for debate."; the RG2 overview bullets.
- **RG3 and Task 3 (about −5):** the overview bullets (C8); the FG-PRM sentence in preliminary results; "We test this rule against rewards for debate wins and for the final decision", moved into Task 3's evaluation as a baseline; "and agreement by chance earns nothing" in the approach (the chance term is a baseline).
- **Challenge paragraphs (about −6):** RG1's "five pieces of evidence" sentence and the token-cost clause; RG2's "keeps the disagreement" clause; RG3's restated citations.
- **Section 3.5 and assorted (about −18):** the peer-review example; the red-team levels and the varied conditions run into their paragraphs; the results list folded into two sentences; Proposition 2's two-agent example tightened; Section 5's first two paragraphs merged.
- **Tables 2–3 (about −4):** shorter testbed B and C cells; two schedule rows shortened to one line.
- **Sections 3.7, 4, 5, 6 (about −9):** Gold Standard Science run into one paragraph; the reporting sentence merged into Section 4's opening; Section 5 without the Paper Digest clause, the list of past area-chair venues, the IJCAI workshop, IQA-EVAL, Liang et al., FaithScore, the survey and the idea generator (all in the CV); Responsibility's six items run in.

**If you will not change the line pitch** (tested): at 14.1 pt ( `\linespread{0.97}` ), dropping Figure 1 as well still leaves about 2 lines over. At 14.5 pt, dropping Figures 1 and 6 as well leaves about 4 lines over.

**Not in the plan:** C11 (same length once you supply the grant's scope) and C13 (12-pt captions and tables would cost about 10 more lines).

## Sources

Papers and pages I opened for this review, beyond those already in the proposal's bibliography:

- Xu (2026), Don't Read the Log · Zhang et al. (2026), Stop Comparing LLM Agents Without Disclosing the Harness · Balasubramanian et al. (2026) · Yang et al. (2026), Co-RL · Hu et al. (2026b) · Denisov-Blanch et al. (2026)
- Chandna, Fluri and Carroll (ICLR 2026 blog track), on Wen et al. (2025)
- Liu et al. (2026), AceMAD · Ebrahimi et al. (2025) · Zheng et al. (2025), CP-WBFT · Xu et al. (2019), On Strategyproof Conference Peer Review · GAVEL (2026) · TIR-Judge (2026)
- Jung et al. (2025), Trust or Escalate · Auzina et al. (2026), ΔBelief-RL · Cai et al. (2025) · Jiang et al. (2026) · Motwani et al. (2024) · Dhull et al. (2022) · Bara (2026)
- Word's single spacing for 12-pt Times New Roman, measured
- Line pitch and font sizes of the four funded examples and of the proposal: measured with pdftotext and pdfplumber on the attached PDFs.
- The NOFO, N0001426SF004 (annotated PDF in the project): formatting and page limit (pp. 15–16), contents (pp. 18–20), evaluation criteria (pp. 29–30), question deadlines (p. 7).